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Head-mounted electronic device with alignment sensors

Abstract

A head-mounted device may have a head-mounted housing. Optical components may be supported by the head-mounted housing. The optical components may include cameras such as front-facing cameras and/or movable optical modules that have displays for displaying images to eye boxes. Sensors may be provided in the head-mounted device to detect changes in orientation between respective optical modules, between respective portions of a chassis, display cover layer, or other head-mounted support structure in the housing, between optical components such as cameras, and/or between optical components and housing structures. Information from these sensors can be used to measure image misalignment such as image misalignment associated with misaligned cameras or misalignment between optical module images and corresponding eye boxes.

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Background/Summary

(1) This application is a continuation of patent application Ser. No. 17/202,198, filed Mar. 15, 2021, which claims the benefit of provisional patent application No. 63/020,630, filed May 6, 2020, both of which are hereby incorporated by reference herein in their entireties.

FIELD

(1) This relates generally to electronic devices, and, more particularly, to electronic devices such as head-mounted devices.

BACKGROUND

(2) Electronic devices such as head-mounted devices may have displays for displaying images and

may have other optical components.

SUMMARY

(3) A head-mounted device may have a head-mounted housing. Optical components may be supported by the head-mounted housing. The optical components may include cameras such as front-facing cameras and/or optical modules that have displays for displaying images to eye boxes.

(4) Sensors may be provided in the head-mounted device to detect changes in position between portions of the head-mounted device. These changes may include changes in the positions between optical modules, between portions of a chassis, display cover layer, or other head-mounted support structure, between optical components such as cameras, and/or between optical components and housing structures. Information from these sensors can be used to measure image misalignment such as image misalignment associated with misaligned cameras or image misalignment between optical module images and the eye boxes to which these images are being provided. During operation of the head-mounted device, images can be warped to compensate for measured misalignment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a top view of an illustrative head-mounted device in accordance with an embodiment.
- (2) FIG. 2 is a rear view of an illustrative head-mounted device in accordance with an embodiment.
- (3) FIG. 3 is a schematic diagram of an illustrative head-mounted device in accordance with an embodiment.
- (4) FIG. 4 is a top view of an illustrative head-mounted device in an undeformed configuration in accordance with an embodiment.
- (5) FIG. 5 is a top view of the illustrative head-mounted device of FIG. 4 in a deformed configuration in accordance with an embodiment.
- (6) FIG. 6 is a side view of an illustrative resistance-based sensor in accordance with an embodiment.
- (7) FIG. 7 is a side view of an illustrative sensor that emits signals and detects reflected signals in accordance with an embodiment.
- (8) FIG. 8 is a side view of an illustrative optical sensor such as a scanning light beam sensor that monitors a pattern in accordance with an embodiment.
- (9) FIGS. 9, 10, and 11 are cross-sectional side views of illustrative strain gauge sensors in accordance with embodiments.
- (10) FIG. 12 is a side view of an illustrative magnetic sensor in accordance with an embodiment.
- (11) FIG. 13 is a cross-sectional side view of an illustrative sensor that uses time-of-flight measurements in accordance with an embodiment.
- (12) FIG. 14 is a cross-sectional side view of an illustrative head-mounted device structure being monitored by a sensor such as a time-of-flight sensor or other sensor in accordance with an embodiment.
- (13) FIG. 15 is a cross-sectional side view of an illustrative sensor arrangement with a light beam that reflects from reflective surfaces in a head-mounted device in accordance with an embodiment.
- (14) FIG. 16 is a diagram of an illustrative optical sensor such as an optical interferometer sensor in accordance with an embodiment.
- (15) FIG. 17 is a diagram of illustrative optical sensing components such as prisms in accordance with an embodiment.
- (16) FIG. 18 is a cross-sectional side view of a portion of a head-mounted device with sensors in accordance with an embodiment.
- (17) FIG. 19 is a perspective view of an illustrative sensor arrangement for an electronic device in

accordance with an embodiment.

(18) FIG. **20** is a top view of an illustrative head-mounted device with sensors in accordance with an embodiment.

(19) FIG. **21** is a flow chart of illustrative operations associated with operating an electronic device in accordance with an embodiment.

DETAILED DESCRIPTION

(20) An electronic device such as a head-mounted device may have a front face that faces away from a user's head and may have an opposing rear face that faces the user's head. Optical modules may be used to provide images to a user's eyes. The positions of the optical modules may be adjusted to accommodate different user interpupillary distances. The head-mounted device may have actuators and optical module guide structures to allow the optical module positions to be adjusted. The head-mounted device may also have other optical components such as front-facing cameras.

(21) A top view of an illustrative head-mounted device is shown in FIG. **1**. As shown in FIG. **1**, head-mounted devices such as electronic device **10** may have head-mounted support structures such as housing **12**. Housing **12** may include portions (e.g., head-mounted support structures **12T**) to allow device **10** to be worn on a user's head. Support structures **12T** may be formed from fabric, polymer, metal, and/or other material. Support structures **12T** may form a strap or other head-mounted support structures to help support device **10** on a user's head. A main support structure (e.g., a head-mounted housing such as main housing portion **12M**) of housing **12** may support electronic components such as displays **14**.

(22) Main housing portion **12M** may include housing structures formed from metal, polymer, glass, ceramic, and/or other material. For example, housing portion **12M** may have housing walls on front face **F** and housing walls on adjacent top, bottom, left, and right side faces that are formed from rigid polymer or other rigid support structures and these rigid walls may optionally be covered with electrical components, fabric, leather, or other soft materials, etc. Housing portion **12M** may also have internal support structures such as a frame and/or structures that perform multiple functions such as controlling airflow and dissipating heat while providing structural support. The walls of housing portion **12M** may enclose internal components **38** in interior region **34** of device **10** and may separate interior region **34** from the environment surrounding device **10** (exterior region **36**). Internal components **38** may include integrated circuits, actuators, batteries, sensors, and/or other circuits and structures for device **10**. Housing **12** may be configured to be worn on a head of a user and may form glasses, a hat, a helmet, goggles, and/or other head-mounted device. Configurations in which housing **12** forms goggles may sometimes be described herein as an example.

(23) Front face **F** of housing **12** may face outwardly away from a user's head and face. Opposing rear face **R** of housing **12** may face the user. Portions of housing **12** (e.g., portions of main housing **12M**) on rear face **R** may form a cover such as cover **12C**. The presence of cover **12C** on rear face **R** may help hide internal housing structures, internal components **38**, and other structures in interior region **34** from view by a user.

(24) Device **10** may have one or more cameras such as cameras **46** of FIG. **1**. For example, forward-facing (front-facing) cameras may allow device **10** to monitor movement of the device **10** relative to the environment surrounding device **10** (e.g., the cameras may be used in forming a visual odometry system or part of a visual inertial odometry system). Forward-facing cameras may also be used to capture images of the environment that are displayed to a user of the device **10**. If desired, images from multiple forward-facing cameras may be merged with each other and/or forward-facing camera content can be merged with computer-generated content for a user.

(25) Device **10** may have any suitable number of cameras **46**. For example, device **10** may have **K** cameras, where the value of **K** is at least one, at least two, at least four, at least six, at least eight, at least ten, at least 12, less than 20, less than 14, less than 12, less than ten, 4-10, or other suitable value. Cameras **46** may be sensitive at infrared wavelengths (e.g., cameras **46** may be infrared

cameras), may be sensitive at visible wavelengths (e.g., cameras **46** may be visible cameras), and/or cameras **46** may be sensitive at other wavelengths. If desired, cameras **46** may be sensitive at both visible and infrared wavelengths.

(26) Cameras **46** that are mounted on front face **F** and that face outwardly (towards the front of device **10** and away from the user) may sometimes be referred to herein as forward-facing or front-facing cameras. Cameras **46** may capture visual odometry information, image information that is processed to locate objects in the user's field of view (e.g., so that virtual content can be registered appropriately relative to real-world objects), image content that is displayed in real time for a user of device **10**, and/or other suitable image data.

(27) Device **10** may have left and right optical modules **40**. Optical modules **40** support electrical and optical components such as light-emitting components and lenses and may therefore sometimes be referred to as optical assemblies, optical systems, optical component support structures, lens and display support structures, electrical component support structures, or housing structures. Each optical module may include a respective display **14**, lens **30**, and support structure such as support structure **32**. Support structure **32**, which may sometimes be referred to as a lens support structure, optical component support structure, optical module support structure, optical module portion, or lens barrel, may include hollow cylindrical structures with open ends or other supporting structures to house displays **14** and lenses **30**. Support structures **32** may, for example, include a left lens barrel that supports a left display **14** and left lens **30** and a right lens barrel that supports a right display **14** and right lens **30**.

(28) Displays **14** may include arrays of pixels or other display devices to produce images. Displays **14** may, for example, include organic light-emitting diode pixels formed on substrates with thin-film circuitry and/or formed on semiconductor substrates, pixels formed from crystalline semiconductor dies, liquid crystal display pixels, scanning display devices, and/or other display devices for producing images.

(29) Lenses **30** may include one or more lens elements for providing image light from displays **14** to respective eyes boxes **13**. Lenses may be implemented using refractive glass lens elements, using mirror lens structures (catadioptric lenses), using Fresnel lenses, using holographic lenses, and/or other lens systems.

(30) When a user's eyes are located in eye boxes **13**, displays (display panels) **14** operate together to form a display for device **10** (e.g., the images provided by respective left and right optical modules **40** may be viewed by the user's eyes in eye boxes **13** so that a stereoscopic image is created for the user). The left image from the left optical module fuses with the right image from a right optical module while the display is viewed by the user.

(31) It may be desirable to monitor the user's eyes while the user's eyes are located in eye boxes **13**. For example, it may be desirable to use a camera to capture images of the user's irises (or other portions of the user's eyes) for user authentication. It may also be desirable to monitor the direction of the user's gaze. Gaze tracking information may be used as a form of user input and/or may be used to determine where, within an image, image content resolution should be locally enhanced in a foveated imaging system. To ensure that device **10** can capture satisfactory eye images while a user's eyes are located in eye boxes **13**, each optical module **40** may be provided with a camera such as camera **42** and one or more light sources such as light-emitting diodes **44** or other light-emitting devices such as lasers, lamps, etc. Cameras **42** and light-emitting diodes **44** may operate at any suitable wavelengths (visible, infrared, and/or ultraviolet). As an example, diodes **44** may emit infrared light that is invisible (or nearly invisible) to the user. This allows eye monitoring operations to be performed continuously without interfering with the user's ability to view images on displays **14**.

(32) Not all users have the same interpupillary distance IPD. To provide device **10** with the ability to adjust the interpupillary spacing between modules **40** along lateral dimension **X** and thereby adjust the spacing IPD between eye boxes **13** to accommodate different user interpupillary

distances, device **10** may be provided with optical module positioning systems in housing **12**. The positioning systems may have guide members and actuators **43** that are used to position optical modules **40** with respect to each other.

(33) Actuators **43** can be manually controlled and/or computer-controlled actuators (e.g., computer-controlled motors) for moving support structures (lens barrels) **32** relative to each other.

Information on the locations of the user's eyes may be gathered using, for example, cameras **42**. The locations of eye boxes **13** can then be adjusted accordingly.

(34) As shown in the rear view of device **10** of FIG. **2**, cover **12C** may cover rear face **R** while leaving lenses **30** of optical modules **40** uncovered (e.g., cover **12C** may have openings that are aligned with and receive modules **40**). As modules **40** are moved relative to each other along dimension **X** to accommodate different interpupillary distances for different users, modules **40** move relative to fixed housing structures such as the walls of main portion **12M** and move relative to each other.

(35) A schematic diagram of an illustrative electronic device such as a head-mounted device or other wearable device is shown in FIG. **3**. Device **10** of FIG. **3** may be operated as a stand-alone device and/or the resources of device **10** may be used to communicate with external electronic equipment. As an example, communications circuitry in device **10** may be used to transmit user input information, sensor information, and/or other information to external electronic devices (e.g., wirelessly or via wired connections). Each of these external devices may include components of the type shown by device **10** of FIG. **3**.

(36) As shown in FIG. **3**, a head-mounted device such as device **10** may include control circuitry **20**. Control circuitry **20** may include storage and processing circuitry for supporting the operation of device **10**. The storage and processing circuitry may include storage such as nonvolatile memory (e.g., flash memory or other electrically-programmable-read-only memory configured to form a solid-state drive), volatile memory (e.g., static or dynamic random-access-memory), etc. Processing circuitry in control circuitry **20** may be used to gather input from sensors and other input devices and may be used to control output devices. The processing circuitry may be based on one or more microprocessors, microcontrollers, digital signal processors, baseband processors and other wireless communications circuits, power management units, audio chips, application specific integrated circuits, etc. During operation, control circuitry **20** may use display(s) **14** and other output devices in providing a user with visual output and other output.

(37) To support communications between device **10** and external equipment, control circuitry **20** may communicate using communications circuitry **22**. Circuitry **22** may include antennas, radio-frequency transceiver circuitry, and other wireless communications circuitry and/or wired communications circuitry. Circuitry **22**, which may sometimes be referred to as control circuitry and/or control and communications circuitry, may support bidirectional wireless communications between device **10** and external equipment (e.g., a companion device such as a computer, cellular telephone, or other electronic device, an accessory such as a point device, computer stylus, or other input device, speakers or other output devices, etc.) over a wireless link. For example, circuitry **22** may include radio-frequency transceiver circuitry such as wireless local area network transceiver circuitry configured to support communications over a wireless local area network link, near-field communications transceiver circuitry configured to support communications over a near-field communications link, cellular telephone transceiver circuitry configured to support communications over a cellular telephone link, or transceiver circuitry configured to support communications over any other suitable wired or wireless communications link. Wireless communications may, for example, be supported over a Bluetooth® link, a WiFi® link, a wireless link operating at a frequency between 10 GHz and 400 GHz, a 60 GHz link, or other millimeter wave link, a cellular telephone link, or other wireless communications link. Device **10** may, if desired, include power circuits for transmitting and/or receiving wired and/or wireless power and may include batteries or other energy storage devices. For example, device **10** may include a coil

and rectifier to receive wireless power that is provided to circuitry in device **10**.

(38) Device **10** may include input-output devices such as devices **24**. Input-output devices **24** may be used in gathering user input, in gathering information on the environment surrounding the user, and/or in providing a user with output. Devices **24** may include one or more displays such as display(s) **14**. Display(s) **14** may include one or more display devices such as organic light-emitting diode display panels (panels with organic light-emitting diode pixels formed on polymer substrates or silicon substrates that contain pixel control circuitry), liquid crystal display panels, microelectromechanical systems displays (e.g., two-dimensional mirror arrays or scanning mirror display devices), display panels having pixel arrays formed from crystalline semiconductor light-emitting diode dies (sometimes referred to as microLEDs), and/or other display devices.

(39) Sensors **16** in input-output devices **24** may include force sensors (e.g., strain gauges, capacitive force sensors, resistive force sensors, etc.), audio sensors such as microphones, touch and/or proximity sensors such as capacitive sensors such as a touch sensor that forms a button, trackpad, or other input device), and other sensors. If desired, sensors **16** may include optical sensors such as optical sensors that emit and detect light, ultrasonic sensors, optical touch sensors, optical proximity sensors, and/or other touch sensors and/or proximity sensors, monochromatic and color ambient light sensors, image sensors (e.g., cameras), fingerprint sensors, iris scanning sensors, retinal scanning sensors, and other biometric sensors, temperature sensors, sensors for measuring three-dimensional non-contact gestures (“air gestures”), pressure sensors, sensors for detecting position, orientation, and/or motion (e.g., accelerometers, magnetic sensors such as compass sensors, gyroscopes, and/or inertial measurement units that contain some or all of these sensors), health sensors such as blood oxygen sensors, heart rate sensors, blood flow sensors, and/or other health sensors, radio-frequency sensors, three-dimensional camera systems such as depth sensors (e.g., structured light sensors and/or depth sensors based on stereo imaging devices that capture three-dimensional images) and/or optical sensors such as self-mixing sensors and light detection and ranging (lidar) sensors that gather time-of-flight measurements (e.g., time-of-flight cameras), humidity sensors, moisture sensors, gaze tracking sensors, electromyography sensors to sense muscle activation, facial sensors, interferometric sensors, time-of-flight sensors, magnetic sensors, resistive sensors, distance sensors, angle sensors, and/or other sensors. In some arrangements, device **10** may use sensors **16** and/or other input-output devices to gather user input. For example, buttons may be used to gather button press input, touch sensors overlapping displays can be used for gathering user touch screen input, touch pads may be used in gathering touch input, microphones may be used for gathering audio input (e.g., voice commands), accelerometers may be used in monitoring when a finger contacts an input surface and may therefore be used to gather finger press input, etc.

(40) If desired, electronic device **10** may include additional components (see, e.g., other devices **18** in input-output devices **24**). The additional components may include haptic output devices, actuators for moving movable housing structures, audio output devices such as speakers, light-emitting diodes for status indicators, light sources such as light-emitting diodes that illuminate portions of a housing and/or display structure, other optical output devices, and/or other circuitry for gathering input and/or providing output. Device **10** may also include a battery or other energy storage device, connector ports for supporting wired communication with ancillary equipment and for receiving wired power, and other circuitry.

(41) It is desirable for optical components in device **10** to remain in satisfactory alignment during operation of device **10**. Consider, as an example, optical modules **40** and front facing cameras **46**. Due to a drop event or other event imparting stress on housing portion **12M**, housing portion **12M** and the components of housing portion **12M** may become deformed and misaligned. For example, the left and right optical modules **40** in device **10** may become misaligned and/or a pair of front-facing cameras **46** may become misaligned. This can adversely affect device performance. For example, if the left and right images being viewed by a user become misaligned with respect to

each other, these images may not fuse satisfactorily in the user's vision and/or a user may experience discomfort during image viewing. Similarly, misalignment of cameras **46** can lead to undesired misalignment between captured images.

(42) To ensure that device **10** is comfortable to wear on a user's head, device **10** should not be too heavy or too large. Particularly when using lightweight and compact structures to form device **10**, however, there is a risk that excessive stress on the structures of device **10** will adversely affect the alignment of optical components of device **10**. To accommodate potential optical component misalignment, device **10** may be provided with sensors that can measure misalignment.

(43) Consider, as an example, head-mounted device **10** of FIG. **4**. In the configuration of FIG. **4**, device **10** has been provided with sensor **16** (e.g., a strain gauge or other sensor). Sensor **16** may, as an example, be coupled to a portion of housing **12M**. During operation, sensor **16** can detect when housing **12M** has become deformed (e.g., bent) due to a drop event or other stress. As shown in FIG. **4**, optical modules **40** provide left and right images to corresponding left and right eye boxes **30** in directions **50** and left and right forward-facing cameras **46** capture corresponding left and right camera images. When housing **12M** of FIG. **4** is not deformed as shown in FIG. **4**, the images provided by optical modules **40** in directions **50** are aligned with eye boxes **30**, so images for the left and right eyes fuse properly into a single image when viewed by the user's left and right eyes. Cameras **46** are also aligned satisfactorily.

(44) When housing **12M** is deformed as shown in FIG. **5**, however, the images provided in directions **50** are no longer aligned with eye boxes **30** and/or the images captured by cameras **46** are no longer aligned satisfactorily. Due to the presence of sensors such as sensor **16**, image misalignment associated with display images and/or camera images can be detected in real time and appropriate compensating action taken. As an example, if it is determined that an image from a given optical module **40** is rotated with respect to a corresponding eye box **30**, device **10** (e.g., control circuitry in device **10** such as control circuitry **20** of FIG. **3**) may apply a corresponding image warping process with a corresponding counterrotation to the image being produced by that optical module. This ensures that the images from optical modules **40** are properly aligned with eye boxes **30** and each other, even if optical modules have become physically misaligned. Similarly, images captured by forward facing cameras **46** may be processed to compensate for detected misalignment between the cameras. Configurations where misalignment between cameras and optical modules and/or other optical component misalignment is detected and compensated may also be used.

(45) One or more sensors may be provided to gather information on the positions of structures in device **10**. These sensors can produce output that is used in detecting misalignment and may therefore sometimes be referred to as misalignment sensors, position sensors, and/or orientation sensors. The sensors may be strain gauges, optical sensors, radio-frequency sensors, acoustic sensors, magnetic sensors, and/or other sensors that detect deformation of housing **12M** and/or other portions of device **10** leading to misalignment of optical components with respect to each other and/or to housing **12**.

(46) Using a misalignment sensor, the orientation of housing **12M** and/or one or more optical components in housing **12M** such as optical modules **40** and/or cameras **46** can be measured. Control circuitry **20** can process the output from one or more orientation sensors in real time and can take suitable action to compensate for the effects of optical component misalignment. As an example, control circuitry **20** can warp or otherwise modify image data (e.g., display output data associated with one or more optical module displays, captured camera images from one or more cameras, etc.) to digitally compensate for misalignment. During image warping, an image is geometrically distorted (e.g., a geometrical image transform is applied to the image such as a shift, shear, rotation, etc.). In this way, device **10** can be operated satisfactorily, even if optical components in device **10** have become misaligned.

(47) Illustrative misalignment sensor configurations are shown in FIGS. **6-20**.

(48) In the example of FIG. 6, misalignment sensor **16** has a first portion **16-1** and a second portion **16-2**. Shaft **16-2'** of portion **16-2** moves in and out of portion **16-1** as the distance **D** between portions **16-1** and **16-2** varies. Sensor **16** may be, for example, a resistance-based sensor (e.g., a potentiometer) that is characterized by a resistance **R** between its terminals **52** that is proportional to distance **D**. Resistance-based sensors and other misalignment sensors **16** may be configured to measure changes in displacement (e.g., distance **D**) and/or changes in angular orientation that are indicative of component movement and/or housing deformation and that are therefore indicative of associated image misalignment.

(49) In the example of FIG. 7, sensor **16** has an emitter **54** and a detector **56** for detecting misalignment between structures in device **10** such as structure **64** and structure **62**. Emitter **54** may emit signals **58** (e.g., light, radio-frequency signals, acoustic signals such as ultrasonic signals, etc.). Emitted signals **58** reflect from a reflective surface of structure in device **10** such as structure **62** (e.g., a portion of an optical module, a housing structure, a portion of a camera, etc.). Emitted signals **58** that have reflected from structure **62** such as reflected signals **60** are detected by detector **56**. The strength of the reflected signals in this example is proportional to distance **D** and can therefore be used by device **10** (e.g., by sensor **16** and/or control circuitry **20**) to measure changes to distance **D** and/or changes in the angle between structures in device **10**.

(50) In the example of FIG. 8, scanning light beam sensor **16** has a patterned target such as target **70**. Sensor **16** of FIG. 8 is used in detecting changes in the angular orientation between structures **80** and **82**. Patterned target **70**, which may sometimes be referred to as a target, pattern, fiducial, etc., may include, for example, a bar code or other recognizable pattern. Sensor **16** may include a light emitter such as scanning light beam emitter **74**. Emitter **74** may be a scanning laser device or other devices that emits a beam of light **76** that is scanned across target **70** in directions **78**. Detector **72** detects corresponding reflected light. Because of the recognizable pattern of target **70**, lateral shifts in the position of target **70** can be detected. Sensor **16** can thereby detect when the angle between structures **80** and **82** has changed (e.g., because structure **80** has shifted laterally in direction **85**).

(51) If desired, a misalignment sensor may include a strain gauge. FIG. 9 is a cross-sectional side view of an illustrative strain gauge formed from conductive trace **84** (e.g., a conductive trace with a serpentine path). Trace **84** is supported by one or more structures in device **10** (e.g., portions of housing **12**, optical modules **40**, cameras **46**, etc.). When trace **84** bends, the effective length of trace **84** and therefore the resistance of trace **84** changes. A Wheatstone bridge or other resistance measurement circuitry that is coupled to the terminals of trace **84** may be used by control circuitry **20** to measure strain.

(52) In the example of FIG. 9, strain gauge metal trace **84** has been formed using a laser activation process. This process, which may sometimes be referred to as laser direct structuring, involves exposing laser activatable material such as material **86** (e.g., polymer with sensitizers or other material that is sensitized to light exposure) to laser light in a desired trace pattern, followed by electrodeposition to grow trace **84** in the exposed areas. Material **86** may be formed from a different material than other structures in device **10** (e.g. material **86** may be a first shot of polymer and supporting material **88** may be a second shot of polymer that forms a portion of housing **12** and/or other structure in a head-mounted device) or a single material may be used in forming these structures. In the example of FIG. 10, strain gauge trace **84** has been formed on layer **90**. Layer **90** may be a layer on structure **88** (e.g., a dielectric coating layer, a layer of light-sensitive polymer, a flexible printed circuit substrate or other substrate that is attached to structure **88** using adhesive, welds, fasteners, and/or other attachment mechanisms, etc.).

(53) FIG. 11 is a cross-sectional side view of an illustrative structure in device **10** (structure **88**) showing how a sensor such as strain gauge may be embedded within the structure. Structure **88** may, as an example, have portions **88M** (e.g., metal portions) and portions **88I** (e.g., insulating portions formed from polymer or other dielectric). Sensor **16** may be a strain gauge formed from

strain gauge trace **84** (e.g., a metal trace) formed in insulator **90** (e.g. a layer of polymer sandwiched between layers **88I**). In general, strain gauge sensors may be embedded within structures such as structure **88** forming housing **12**, forming optical component packages (e.g., portions of optical modules **40**, portions of cameras **46**), and/or other structures in device **10**.

(54) FIG. **12** is a cross-sectional side view of an illustrative magnetic sensor. Sensor **16** includes a permanent magnet such as magnet **94** and a magnetic sensing device such as a Hall effect sensor or other sensor that can measure magnetic fields (e.g. magnetic sensor **92**). The measured magnetic field strength at sensor **92** is a function of distance D between magnet **94** and sensor **92**, thereby allowing sensor **16** to measure distance D (as an example).

(55) Time-of-flight measurements may be made using sensor **16**, if desired. Consider, as an example, sensor **16** of FIG. **13**. Sensor **16** of FIG. **13** includes a signal emitter such as source **96** and a signal sensor such as detector **98**. During operation, source **96** emits signal **102** (e.g., a pulse) and detects reflected signal **104** (e.g., the pulse following reflection of the pulse from the surface of structure **110** in device **10** (e.g., a portion of support structure **12**, optical modules **40**, cameras **46**, or other structure in device **10**). The emitted signal from source **96** may be an optical signal (light), an acoustic signal (sound such as an ultrasonic acoustic signal), or a radio-frequency signal (e.g., a signal having a frequency of 1 MHz-100 GHz or other suitable radio frequency). Detector **98** (e.g., a photodetector sensitive to light, a microphone sensitive to sound, or an antenna sensitive to radio-frequency signals) measures the time at which reflected signal **104** is received relative to the time at which source **96** emitted signal **102**. Using the known speed of propagation of signals **102** and **104** and the measured time of flight of the emitted signal (e.g., the time between pulse transmission and pulse reception), sensor **16** and/or control circuitry **20** can determine the distance D between sensor **16** and structure **100**.

(56) If desired, a front-facing display and/or other components (e.g., a touch sensor layer) may be provided on front face **F**. For example, housing portion **12M** may include a display cover layer that forms a front member that covers front face **F**. As shown in FIG. **14**, a display cover layer (e.g., a display cover layer with a curved cross-sectional profile or other suitable shape) such as display cover layer **106** (e.g., a front portion of housing portion **12M**) may overlap a pixel array such as pixel array **108** (sometimes referred to as a display, display layer, display panel, etc.). Pixel array **108** may be a liquid crystal display panel, an organic light-emitting display panel, a display panel with other light-emitting diodes, etc.

(57) Display cover layer **106** may be formed from a layer of glass, clear polymer, or other transparent material that allows pixel array **108** to be viewed through display cover layer **106**. During operation, pixel array **108** may be used in presenting images on front face **F** (e.g., images that are viewable by the user when device **10** is not being worn on the user's head and images that are viewable by nearby people when device **10** is being worn on the user's head). To detect bending of display cover layer **106**, which may cause cameras **46** or other optical components mounted on or adjacent to display cover layer **106** to become misaligned with respect to each other or which may be indicative of bending of housing portion **12M** that causes optical modules **40** to become misaligned with respect to each other and/or with respect to cameras **46**, display cover layer **106** may be provided with one or more sensors such as sensor **16**.

(58) Sensor **16** on display cover layer **106** of FIG. **14** may be, for example, a time-of-flight sensor having a signal emitter such as source **96** and a signal detector such as detector **98**. During measurement operations, control circuitry **20** may use source **96** to emit signal **102**. Signal **102** may reflect from reflector **110** as reflected signal **104**, which is detected by detector **98**. Reflector **110** may be a structure in housing portion **12M** such as a portion of display cover layer **106** or a mirror or other structure that is attached to display cover layer **106**. The time of flight of the signal from source **96** to reflector **110** and back to detector **98** may be used to compute distance D (e.g., the distance separation different portions of housing portion **12M** such as different left and right side portions or top and bottom portions of display cover layer **106**). If bending or other deformation of

display cover layer **106** that is associated with image misalignment is detected, compensating adjustments may be made by control circuitry **20**.

(59) In the illustrative configuration of FIG. **15**, sensor **16** has a light source such as source **112**. Light source **112** may be, for example, a light-emitting diode or a laser that emits a beam of light such as light beam **118**. Device **10** may be configured so that light beam **118** reflects from multiple structures in housing portion **12M** (e.g., multiple reflectors **116**) before being detected by detector **114** of sensor **16**. Structures **116** may be mirrors or reflective portions of a chassis or other structural support in housing portion **12M**, reflective portions of optical modules **40**, a reflective rear portion of a display cover layer or other structure on front face **F** of device **10**, and/or other optical component structures and/or housing structures in device **10**. Detector **114** may have an array of photodetectors (e.g., a one-dimensional or two-dimensional array of photodiodes), allowing detector **114** to detect the position at which light beam **118** strikes detector **114**. During operation, deformation of housing portion **12M** and/or components in device **10** (e.g., changes in the positions of optical modules **40** in the example of FIG. **15**) may result in image misalignment. This misalignment may be detected by detecting changes in the position at which light beam **118** illuminates the photodetector array of sensor **114**.

(60) If desired, interferometry (e.g., optical interferometry) may be used in measuring misalignment. Consider, as an example, the interferometer of FIG. **16**. As shown in FIG. **16**, laser interferometer sensor **16** of FIG. **16** may have a coherent light source such as laser **120**. A first portion of the light emitted by laser **120** may pass through beam splitter **130**, may travel along first interferometer arm **126**, may reflect from structure **128**, may reflect from beam splitter **130**, may travel along second interferometer arm **122**, may reflect from structure **124**, and may pass through beam splitter **130** before reaching light detector **134**. A second portion of the light emitted by light source **120** may reflect from beam splitter **130** toward light detector **114**. Changes in distance **D** affect the length of interferometer arm **126** and therefore affect the phase of the first portion of the light reaching detector **134**. As a result of interference between the first and second portions of the coherent light reaching detector **134**, the relative phase of the first and second portions of light (and therefore changes in distance **D**) can be measured by sensor **16**. Structure **128** may be a portion of housing **12M**, a portion of an optical module or other optical component, or other structure in device **10** from which it is desired to gather a distance measurement.

(61) If desired, a single light source (e.g. laser **112** of FIG. **16**) may be using in making multiple optical measurements (e.g., multiple time-of-flight measurements, multiple light beam angle measurements of the type described in connection with FIG. **15**, multiple interferometric measurements, etc.). Optical couplers (e.g., beam splitters, prisms, gratings, etc.) may be used in dividing a single light beam from laser **112** into multiple light beams to use in one or more of these measurements, thereby reducing the number of light sources used in device **10**. An illustrative configuration in which three prisms **140** are being used to split a single beam from laser **112** into three respective misalignment measurement beams is shown in FIG. **17**.

(62) As described in connection with FIG. **14**, front-facing cameras **46** and/or other optical components may be mounted on or adjacent to a portion of housing portion **12M** such as display cover layer **106**. This may help align the front-facing cameras or other optical components. To detect misalignment of images being captured by cameras **46**, sensors **16** may be coupled to display cover layer **106**. As shown in FIG. **18**, for example, front-facing cameras **46** may be mounted to the inner surface of display cover layer **106** and may operate through display cover layer **106**. Placing cameras **46** against display cover layer **106** in this way may help align cameras **46** (e.g., left and right forward-facing cameras in device **10**) and may help align the images produced by these cameras. Sensors **16** may be strain gauges formed from metal traces on the inner surface of display cover layer **106** or may be strain gauges attached to the inner surface of display cover layer **106** by adhesive (as an example). In the event that display cover layer **106** deforms during operation of device **10**, the misalignment of the images being captured by cameras **46** can be detected in real

time. This allows compensating actions to be taken on the misaligned images (e.g., image warping operations on the image data may be performed). Sensors **16** may be attached to the inner surface of display cover layer **106** between each pair of respective cameras **46** and/or at other locations. If desired, deformation of housing portion **12M** such as deformation of display cover layer **106** may be used in compensating for optical module misalignment (e.g., in arrangements in which bending of display cover layer **106** is associated with misalignment between the images from modules **40** and eye boxes **30**).

(63) If desired, multiple sensors **16** may be used to measure the orientation between optical components and/or other structures in device **10** (e.g., by triangulation). Consider, as an example, the scenario of FIG. **19**. In this example, optical modules **40** include left and right modules. The orientation between a right-facing surface of a left optical module (surface LH) and a corresponding left-facing surface of a right optical module (surface RH) is being measured using three sensors **16** (e.g., three distance sensors such as optical sensors or other sensors **16** of the types described in connection with FIGS. **4-18**). For example, a first distance sensor may measure a first distance L1 between point **142** on surface LH and corresponding point **144** on surface RH, a second distance sensor may measure a second distance L2 between point **146** on surface LH and corresponding point **148** on surface RH, and a third distance sensor may measure a third distance L3 between point **150** on surface LH and corresponding point **152** on surface RH. By using these three distance measurements (L1, L2, and L3), the relative orientation in X, Y, and Z (including any relative tilt about each of these axes and any changes in position) between the left and right optical modules **40** (and/or any other pair of structures in device **10**) can be determined.

(64) In general, the relative positions (including displacement and/or angle) between any pair of structures in device **10** can be measured using one or more sensors **16**. Consider, as an example, device **10** of FIG. **20**. As shown in FIG. **20**, sensors **16** such as strain gauges on front face F of housing portion **12M** and/or other structures in device **10** may measure image misalignment associated with images captured by forward facing cameras **46**. Sensors **16** may, for example, measure distance D1 (and/or orientation angle) between points **154** and **156** on display cover layer **106** (or other portion of housing portion **12M**). Additional sensors **16** may measure distances (and/or orientation angles) such as distance D2 (and/or angle) between point **158** (e.g., a portion of a rigid frame in cover **12C**, a chassis structure such as a rigid frame in housing portion **12M**, a left housing wall, etc.) and point **160** (e.g., a portion of lens barrel **32** or other optical module structure for a left optical module **40**), distance D2' (and/or angle) between point **162** (e.g., a portion of left optical module **40**) and point **164** (e.g., a portion of housing portion **12M** such as a portion of display cover layer **106**), distance D3 (and/or angle) between point **166** (e.g., a portion of a left optical module **40**) and point **168** (e.g., a portion of a right optical module **40**), and/or other distances and/or angles between respective portions of device **10**. As these examples demonstrate, one or more sensors **16** may be used in measuring the relative orientation between a head-mounted support structure of device **10** (e.g., a chassis or other device housing structure) and an optical component such as optical module **40**, between a pair of optical modules **40**, and/or between different portions of a head-mounted support structure. In configurations in which left and right forward-facing cameras **46** are mounted to respective left and right optical modules **40** (e.g., in positions **46'** of FIG. **20**), measuring the relative orientation between optical modules **40** serves to measure the relative orientation between the forward-facing cameras. In configurations in which forward-facing cameras **46** are mounted elsewhere in device **10**, sensors **16** can measure the relative position between cameras **46**, the position between each camera **46** and housing portion **12M**, and/or the relative position between each camera **46** and each optical module **40** (as examples).

(65) Misalignment between respective cameras (and the images captured by the cameras) between a given camera and a given optical module (and their associated images), and/or between respective optical modules may be measured directly (e.g., by an associated sensor) or indirectly

(e.g., by using a first sensor to detect a first misalignment such as a misalignment between a first component and housing portion **12M** and to detect a second misalignment such as a misalignment between housing portion **12M** and a second component, thereby producing misalignment information corresponding to misalignment between the first and second components).

(66) Illustrative operations involved in operating device **10** are shown in FIG. **21**.

(67) During the operations of block **170**, sensors **16** may be used to gather information on the orientation of optical components in device **10**. Each optical component (e.g., each camera **46**, each optical module **40**, etc.) may have a respective sensor **16** and/or sensors **16** may measure support structure deformation that is associated with changes in the orientations of the optical components. This allows optical-component-to-optical-component misalignment to be monitored. If desired, structures in housing portion **12M** (e.g., a display cover layer, a frame or other chassis structure, and/or other optical component support structures, etc.) may be provided with one or more sensors **16** and/or one or more sensors **16** may be used in measuring optical-component-to-support-structure orientation. In this way, optical-component-to-optical component misalignment and corresponding image misalignment may be measured directly or indirectly. Position measurements may be associated with camera-to-camera alignment, camera-to-optical-module alignment, camera-to-housing alignment, optical module-to-housing alignment, optical-module-to-optical-module alignment, and/or housing-to-housing alignment and each of these measurements may be associated with potential misalignment between images, between an image and an eye box, etc.

(68) During the operations of block **172**, control circuitry **20** may process image data (captured images from cameras **46** and/or displayed images on optical modules **40**) to compensate for misalignment measured by sensors **16** during the operations of block **170**. For example, if it is determined that an optical module (e.g., a right-hand optical module) is presenting an image that has become rotated counterclockwise by angle A relative to an eye box, compensating image warping may be performed on the image to rotate the image clockwise by angle A and thereby compensate for the misalignment. Camera images may likewise be warped. The image warping transforms that are applied during misalignment compensation operations may include geometrical transforms such as shifts, shears, rotations, etc.

(69) Consider, as an example, a scenario in which sensors **16** determine that first and second optical components (e.g., a pair of cameras **46**, a pair of modules **40**, a camera and a module, etc.) are misaligned with respect to each other. During the image compensation operations of block **172**, control circuitry **20** may, based on the measured misalignment, warp a first image captured by or displayed by the first optical component and/or may warp a second image captured by or displayed by a second optical component, thereby compensating the images for the measured misalignment.

(70) As shown by line **174**, the optical component orientation measurements of block **172** to detect component and image misalignment and the corresponding misalignment compensation image processing adjustments that are performed at block **172** may be performed continuously (e.g., upon detection of a drop event, upon power up, in response to a user-initiated calibration sequence during which a user is asked to move device **10** so that sensors **60** can detect misalignment, in accordance with a predetermined schedule, etc.).

(71) As described above, one aspect of the present technology is the gathering and use of information such as information from input-output devices. The present disclosure contemplates that in some instances, data may be gathered that includes personal information data that uniquely identifies or can be used to contact or locate a specific person. Such personal information data can include demographic data, location-based data, telephone numbers, email addresses, twitter ID's, home addresses, data or records relating to a user's health or level of fitness (e.g., vital signs measurements, medication information, exercise information), date of birth, username, password, biometric information, or any other identifying or personal information.

(72) The present disclosure recognizes that the use of such personal information, in the present

technology, can be used to the benefit of users. For example, the personal information data can be used to deliver targeted content that is of greater interest to the user. Accordingly, use of such personal information data enables users to calculated control of the delivered content. Further, other uses for personal information data that benefit the user are also contemplated by the present disclosure. For instance, health and fitness data may be used to provide insights into a user's general wellness, or may be used as positive feedback to individuals using technology to pursue wellness goals.

(73) The present disclosure contemplates that the entities responsible for the collection, analysis, disclosure, transfer, storage, or other use of such personal information data will comply with well-established privacy policies and/or privacy practices. In particular, such entities should implement and consistently use privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining personal information data private and secure. Such policies should be easily accessible by users, and should be updated as the collection and/or use of data changes. Personal information from users should be collected for legitimate and reasonable uses of the entity and not shared or sold outside of those legitimate uses. Further, such collection/sharing should occur after receiving the informed consent of the users. Additionally, such entities should consider taking any needed steps for safeguarding and securing access to such personal information data and ensuring that others with access to the personal information data adhere to their privacy policies and procedures. Further, such entities can subject themselves to evaluation by third parties to certify their adherence to widely accepted privacy policies and practices. In addition, policies and practices should be adapted for the particular types of personal information data being collected and/or accessed and adapted to applicable laws and standards, including jurisdiction-specific considerations. For instance, in the United States, collection of or access to certain health data may be governed by federal and/or state laws, such as the Health Insurance Portability and Accountability Act (HIPAA), whereas health data in other countries may be subject to other regulations and policies and should be handled accordingly. Hence different privacy practices should be maintained for different personal data types in each country.

(74) Despite the foregoing, the present disclosure also contemplates embodiments in which users selectively block the use of, or access to, personal information data. That is, the present disclosure contemplates that hardware and/or software elements can be provided to prevent or block access to such personal information data. For example, the present technology can be configured to allow users to select to “opt in” or “opt out” of participation in the collection of personal information data during registration for services or anytime thereafter. In another example, users can select not to provide certain types of user data. In yet another example, users can select to limit the length of time user-specific data is maintained. In addition to providing “opt in” and “opt out” options, the present disclosure contemplates providing notifications relating to the access or use of personal information. For instance, a user may be notified upon downloading an application (“app”) that their personal information data will be accessed and then reminded again just before personal information data is accessed by the app.

(75) Moreover, it is the intent of the present disclosure that personal information data should be managed and handled in a way to minimize risks of unintentional or unauthorized access or use. Risk can be minimized by limiting the collection of data and deleting data once it is no longer needed. In addition, and when applicable, including in certain health related applications, data de-identification can be used to protect a user's privacy. De-identification may be facilitated, when appropriate, by removing specific identifiers (e.g., date of birth, etc.), controlling the amount or specificity of data stored (e.g., collecting location data at a city level rather than at an address level), controlling how data is stored (e.g., aggregating data across users), and/or other methods.

(76) Therefore, although the present disclosure broadly covers use of information that may include personal information data to implement one or more various disclosed embodiments, the present

disclosure also contemplates that the various embodiments can also be implemented without the need for accessing personal information data. That is, the various embodiments of the present technology are not rendered inoperable due to the lack of all or a portion of such personal information data.

(77) Physical environment: A physical environment refers to a physical world that people can sense and/or interact with without aid of electronic systems. Physical environments, such as a physical park, include physical articles, such as physical trees, physical buildings, and physical people. People can directly sense and/or interact with the physical environment, such as through sight, touch, hearing, taste, and smell.

(78) Computer-generated reality: in contrast, a computer-generated reality (CGR) environment refers to a wholly or partially simulated environment that people sense and/or interact with via an electronic system. In CGR, a subset of a person's physical motions, or representations thereof, are tracked, and, in response, one or more characteristics of one or more virtual objects simulated in the CGR environment are adjusted in a manner that comports with at least one law of physics. For example, a CGR system may detect a person's head turning and, in response, adjust graphical content and an acoustic field presented to the person in a manner similar to how such views and sounds would change in a physical environment. In some situations (e.g., for accessibility reasons), adjustments to characteristic(s) of virtual object(s) in a CGR environment may be made in response to representations of physical motions (e.g., vocal commands). A person may sense and/or interact with a CGR object using any one of their senses, including sight, sound, touch, taste, and smell. For example, a person may sense and/or interact with audio objects that create 3D or spatial audio environment that provides the perception of point audio sources in 3D space. In another example, audio objects may enable audio transparency, which selectively incorporates ambient sounds from the physical environment with or without computer-generated audio. In some CGR environments, a person may sense and/or interact only with audio objects. Examples of CGR include virtual reality and mixed reality.

(79) Virtual reality: A virtual reality (VR) environment refers to a simulated environment that is designed to be based entirely on computer-generated sensory inputs for one or more senses. A VR environment comprises a plurality of virtual objects with which a person may sense and/or interact. For example, computer-generated imagery of trees, buildings, and avatars representing people are examples of virtual objects. A person may sense and/or interact with virtual objects in the VR environment through a simulation of the person's presence within the computer-generated environment, and/or through a simulation of a subset of the person's physical movements within the computer-generated environment.

(80) Mixed reality: In contrast to a VR environment, which is designed to be based entirely on computer-generated sensory inputs, a mixed reality (MR) environment refers to a simulated environment that is designed to incorporate sensory inputs from the physical environment, or a representation thereof, in addition to including computer-generated sensory inputs (e.g., virtual objects). On a virtuality continuum, a mixed reality environment is anywhere between, but not including, a wholly physical environment at one end and virtual reality environment at the other end. In some MR environments, computer-generated sensory inputs may respond to changes in sensory inputs from the physical environment. Also, some electronic systems for presenting an MR environment may track location and/or orientation with respect to the physical environment to enable virtual objects to interact with real objects (that is, physical articles from the physical environment or representations thereof). For example, a system may account for movements so that a virtual tree appears stationary with respect to the physical ground. Examples of mixed realities include augmented reality and augmented virtuality. Augmented reality: an augmented reality (AR) environment refers to a simulated environment in which one or more virtual objects are superimposed over a physical environment, or a representation thereof. For example, an electronic system for presenting an AR environment may have a transparent or translucent display through

which a person may directly view the physical environment. The system may be configured to present virtual objects on the transparent or translucent display, so that a person, using the system, perceives the virtual objects superimposed over the physical environment. Alternatively, a system may have an opaque display and one or more imaging sensors that capture images or video of the physical environment, which are representations of the physical environment. The system composites the images or video with virtual objects, and presents the composition on the opaque display. A person, using the system, indirectly views the physical environment by way of the images or video of the physical environment, and perceives the virtual objects superimposed over the physical environment. As used herein, a video of the physical environment shown on an opaque display is called “pass-through video,” meaning a system uses one or more image sensor(s) to capture images of the physical environment, and uses those images in presenting the AR environment on the opaque display. Further alternatively, a system may have a projection system that projects virtual objects into the physical environment, for example, as a hologram or on a physical surface, so that a person, using the system, perceives the virtual objects superimposed over the physical environment. An augmented reality environment also refers to a simulated environment in which a representation of a physical environment is transformed by computer-generated sensory information. For example, in providing pass-through video, a system may transform one or more sensor images to impose a select perspective (e.g., viewpoint) different than the perspective captured by the imaging sensors. As another example, a representation of a physical environment may be transformed by graphically modifying (e.g., enlarging) portions thereof, such that the modified portion may be representative but not photorealistic versions of the originally captured images. As a further example, a representation of a physical environment may be transformed by graphically eliminating or obfuscating portions thereof. Augmented virtuality: an augmented virtuality (AV) environment refers to a simulated environment in which a virtual or computer generated environment incorporates one or more sensory inputs from the physical environment. The sensory inputs may be representations of one or more characteristics of the physical environment. For example, an AV park may have virtual trees and virtual buildings, but people with faces photorealistically reproduced from images taken of physical people. As another example, a virtual object may adopt a shape or color of a physical article imaged by one or more imaging sensors. As a further example, a virtual object may adopt shadows consistent with the position of the sun in the physical environment.

(81) Hardware: there are many different types of electronic systems that enable a person to sense and/or interact with various CGR environments. Examples include head mounted systems, projection-based systems, heads-up displays (HUDs), vehicle windshields having integrated display capability, windows having integrated display capability, displays formed as lenses designed to be placed on a person's eyes (e.g., similar to contact lenses), headphones/earphones, speaker arrays, input systems (e.g., wearable or handheld controllers with or without haptic feedback), smartphones, tablets, and desktop/laptop computers. A head mounted system may have one or more speaker(s) and an integrated opaque display. Alternatively, a head mounted system may be configured to accept an external opaque display (e.g., a smartphone). The head mounted system may incorporate one or more imaging sensors to capture images or video of the physical environment, and/or one or more microphones to capture audio of the physical environment. Rather than an opaque display, a head mounted system may have a transparent or translucent display. The transparent or translucent display may have a medium through which light representative of images is directed to a person's eyes. The display may utilize digital light projection, OLEDs, LEDs, pLEDs, liquid crystal on silicon, laser scanning light sources, or any combination of these technologies. The medium may be an optical waveguide, a hologram medium, an optical combiner, an optical reflector, or any combination thereof. In one embodiment, the transparent or translucent display may be configured to become opaque selectively. Projection-based systems may employ retinal projection technology that projects graphical images onto a person's retina. Projection

systems also may be configured to project virtual objects into the physical environment, for example, as a hologram or on a physical surface.

(82) The foregoing is merely illustrative and various modifications can be made to the described embodiments. The foregoing embodiments may be implemented individually or in any combination.

Claims

1. A head-mounted device, comprising: a head-mounted housing; a camera supported by the head-mounted housing; a display supported by the head-mounted housing and configured to display an image; and a sensor configured to measure misalignment of at least one of the camera or the display, wherein the displayed image is compensated for the measured misalignment and wherein the sensor is selected from the group consisting of: a resistance-based sensor that measures a deformation of the head-mounted housing; an emitter that illuminates a target and a detector that detects light reflected from the target; and an optical interferometer having a beam splitter that splits light into first and second light beams, wherein the optical interferometer is configured to measure a relative phase of the first and second light beams.
2. The head-mounted device defined in claim 1 further comprising control circuitry configured to apply image warping to the displayed image to compensate for the measured misalignment.
3. The head-mounted device defined in claim 1 wherein the displayed image comprises an image captured by the camera.
4. The head-mounted device defined in claim 1 wherein the camera is configured to capture images of real-world objects, wherein the displayed images comprise virtual image content that is registered to the real-world objects based on the captured images.
5. A head-mounted device, comprising: a head-mounted housing; a camera supported by the head-mounted housing; a display supported by the head-mounted housing and configured to display an image; a sensor configured to measure misalignment of at least one of the camera or the display, wherein the displayed image is compensated for the measured misalignment; and an additional display supported by the head-mounted housing, wherein the display and the additional display face opposite directions.
6. The head-mounted device defined in claim 5 wherein the sensor is selected from the group consisting of: a strain gauge, an optical sensor, a radio-frequency sensor, an acoustic sensor, an orientation sensor, and a magnetic sensor.
7. The head-mounted device defined in claim 5 further comprising a display cover layer overlapping the additional display, wherein the sensor is coupled to the display cover layer and is configured to measure deformation of the display cover layer.
8. A head-mounted device, comprising: a head-mounted housing; a camera supported by the head-mounted housing; a display supported by the head-mounted housing; a transparent housing layer on which images from the display are viewable; and a sensor coupled to the transparent housing layer and configured to detect deformation of the transparent housing layer, wherein at least one of the camera or the display are adjusted to compensate for deformation of the transparent housing layer, and wherein the camera captures images through the transparent housing layer.
9. The head-mounted device defined in claim 8 wherein the transparent housing layer comprises a display cover layer.
10. The head-mounted device defined in claim 8 wherein the captured images are compensated based on the deformation of the transparent housing layer.
11. A head-mounted device, comprising: a head-mounted housing; a camera supported by the head-mounted housing; a display supported by the head-mounted housing; a transparent housing layer on which images from the display are viewable; and a sensor coupled to the transparent housing layer and configured to detect deformation of the transparent housing layer, wherein at least one of the

- camera or the display are adjusted to compensate for deformation of the transparent housing layer, and wherein the sensor comprises a strain gauge.
12. A head-mounted device, comprising: a head-mounted housing; a display supported by the head-mounted housing; a transparent housing layer on which image content from the display is viewable; first and second forward-facing cameras that capture images through the transparent housing layer; and a sensor configured to detect misalignment between the first and second forward-facing cameras, wherein at least one of the displayed image content or the captured images is compensated based on the measured misalignment.
13. The head-mounted device defined in claim 12 wherein the sensor comprises a strain gauge configured to measure a distance between a first portion of the transparent housing layer and a second portion of the transparent housing layer.
14. The head-mounted device defined in claim 12 further comprising control circuitry configured to apply image warping to the captured images to compensate for the measured misalignment.
15. The head-mounted device defined in claim 12 further comprising an additional display supported by the head-mounted housing, wherein the display and the additional display face opposite directions.
16. The head-mounted device defined in claim 15 wherein the additional display is configured to display virtual image content that is registered to real-world objects based on the captured images.
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